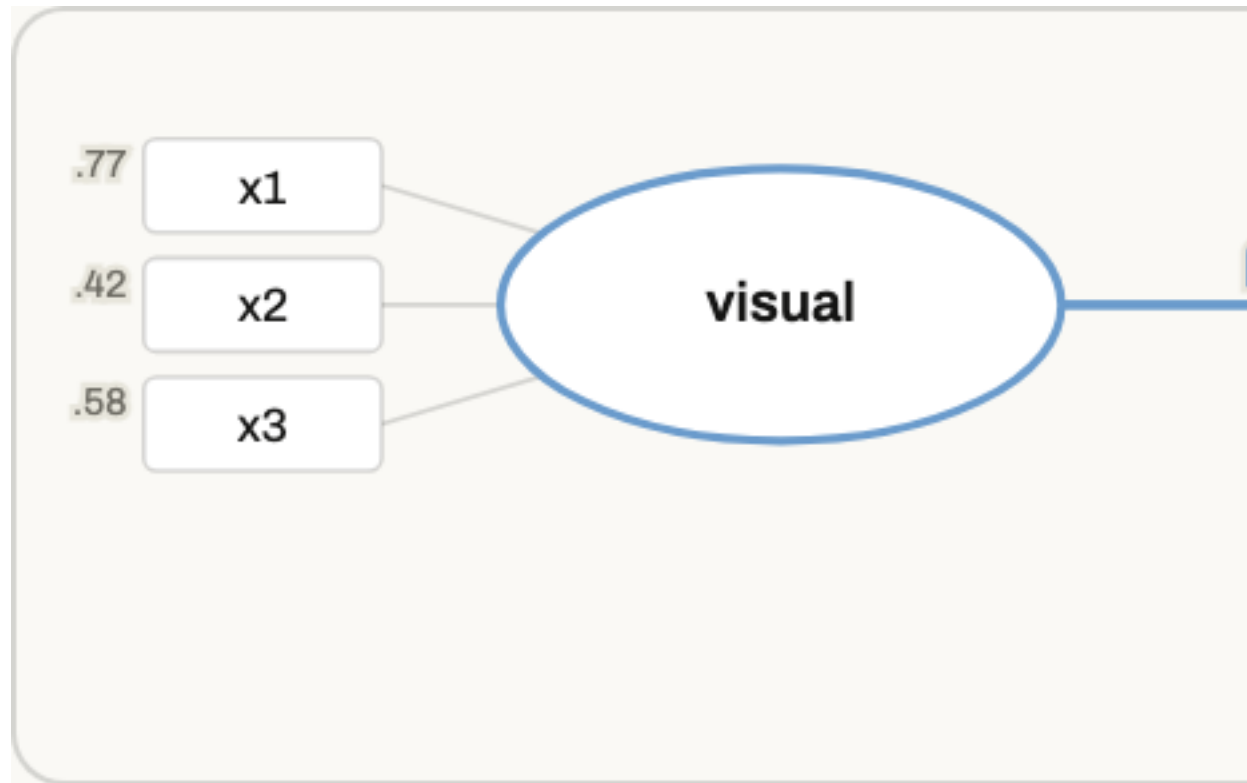


1 CB-SEM

Construct / Item	Mean	SD	B	SE	z	p	Std. loading	ω	α	CR	AVE
<i>visual</i>								<i>.61</i>	<i>.63</i>	<i>.61</i>	<i>.37</i>
x1	4.94	1.17	1.00	0.00	0	<.001	.77				
x2	6.09	1.18	0.55	0.14	3.97	<.001	.42				
x3	2.25	1.13	0.73	0.15	4.85	<.001	.58				
<i>textual</i>								<i>.89</i>	<i>.88</i>	<i>.89</i>	<i>.72</i>
x4	3.06	1.16	1.00	0.00	0	<.001	.85				
x5	4.34	1.29	1.11	0.07	16.50	<.001	.86				
x6	2.19	1.10	0.93	0.06	14.80	<.001	.84				
<i>speed</i>								<i>.69</i>	<i>.69</i>	<i>.69</i>	<i>.42</i>
x7	4.19	1.09	1.00	0.00	0	<.001	.57				
x8	5.53	1.01	1.18	0.15	7.90	<.001	.72				
x9	5.37	1.01	1.08	0.38	2.82	.005	.67				
	visual	textual	speed								
visual	.61			<i>*p<.05, **p<.01, ***p<.001</i>							
textual	.46***	.85									
speed	.47***	.28***	.65								
	visual	textual	speed								
visual											
textual	.38										
speed	.39	.28									
χ^2 (df, p)		χ^2 /df	CFI	TLI	RMSEA [90% CI]		SRMR				
85.31 (24, <.001)		3.55	.93	.90	.09 [.07, .11]		.07				
H	Path		B	Std. β	p	Percentile 95% CI		BC 95% CI		Result	
						Lower	Upper	Lower	Upper		
<i>Direct paths</i>											
H1	visual $\rightarrow$ textual		0.50	.46	<.001	.32	.70	.33	.72	Support	
H2	textual $\rightarrow$ speed		0.05	.09	.333	-.05	.17	-.04	.17	Not sup	
H3	visual $\rightarrow$ speed		0.30	.43	<.001	.14	.45	.16	.48	Support	
<i>Indirect effects</i>											
H4	visual $\rightarrow$ textual $\rightarrow$ speed		0.03	.04	.346	-.03	.09	-.02	.10	Not sup	



First confirm the measurement model (loadings high, CR/AVE adequate) and overall fit indices. Then read the structural paths: each std. β with p/CI is a hypothesized relationship between constructs; R^2 shows variance explained in each outcome construct. CB-SEM is confirmatory: the measurement model must be specified from theory a priori — any post-hoc respecification (e.g. from modification indices) is exploratory, must be reported as such, and ideally cross-validated on a fresh sample.

The model fit well (CFI=.93, RMSEA=.09, SRMR=.07); the path from visual to textual gave β =.46, $p < .001$.

Statistical basis: Model estimation (lavaan) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." *Journal of Statistical Software*, 48(2), 1-36.); Fit-index cutoffs (CFI/TLI $\geq .95$, RMSEA $\leq .06$, SRMR $\leq .08$) (Hu, L., Bentler, P. M. (1999). "Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives." *Structural Equation Modeling*, 6(1), 1-55.); Cutoffs are guidelines, not pass/fail gates (Marsh, H. W., Hau, K. T., Wen, Z. (2004). "In search of golden rules: Comment on hypothesis-testing approaches to setting cutoff values for fit indices." *Structural Equation Modeling*, 11(3), 320-341.); RMSEA 90% confidence interval (MacCallum, R. C., Browne, M. W., Sugawara, H. M. (1996). "Power analysis and determination of sample size for covariance structure modeling." *Psychological Methods*, 1(2), 130-149.); AVE / Fornell-Larcker discriminant

validity (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." *Journal of Marketing Research*, 18(1), 39-50.); Reliability preference (ω over α) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." *Psychological Methods*, 23(3), 412-433.); Reporting standard (APA JARS-Quant) (Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., Rao, S. M. (2018). "Journal article reporting standards for quantitative research in psychology." *American Psychologist*, 73(1), 3-25.); Double-mean-centering for latent moderation (semTools::indProd) (Lin, G. C., Wen, Z., Marsh, H. W., Lin, H. S. (2010). "Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies." *Structural Equation Modeling*, 17(3), 374-391.); Mean-centering strategy for latent interactions (precursor) (Marsh, H. W., Wen, Z., Hau, K. T. (2004). "Structural equation models of latent interactions: evaluation of alternative estimation strategies and indicator construction." *Psychological Methods*, 9(3), 275-300.); Simple slopes at -1SD/mean/+1SD (Aiken, L. S., West, S. G. (1991). *Multiple Regression: Testing and Interpreting Interactions*. Sage.).